Petrographic Analysis of the Apollo Lunar Thin Section 10047-r2

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1 Introduction

The lunar rock 10047 was returned by the Apollo 11 mission from Mare Tranquillitatis in 1969. Paired rotational series of PPL and XPL images of sample 10047-r2 at 5° intervals was acquired from Virtual Microscope [1] and subsequently aligned to ensure a consistent field of view.

2 Petrographic Observations

2.1 Pale brown grains

In PPL, the thin section is dominated by two contrasting phases. Large, pale brown crystals with abundant cleavage (faint parallel striations), and exhibit no pleochroism. In XPL, these grains appear moderate inference colors, in warm browns and oranges. The central grain is conspicuously large, spanning a substantial fraction of the field of view. Although appearing as a single large crystal in PPL, the central grain exhibits non-uniform extinction and interference colors during stage rotation. There's a patchwork of slightly different browns and oranges, with some darker, more olive-toned regions (for instance, toward the lower part of the crystal and near some of the margins), grading into lighter golden-brown areas. The irregular, patchy color variations in interference color across individual crystals during stage rotation likely reflect compositional zoning, consistent with changes in melt composition during crystallization [3].

2.2 Lath-shaped low-relief grains

Surrounding and interpenetrating the pale brown grains are bright, low-relief crystals forming elongate laths and irregular interstitial patches. They stand out in XPL, displaying the black-and-white polysynthetic twinning in gray and white first-order tones. The twinned laths radiate around and project into the pale brown grains. Their bright appearance in PPL and lath-shaped outlines with twinning identify them as plagioclase feldspar. Several thin plagioclase laths can be seen partially or completely enclosed within the larger pale brown pyroxene crystals.

2.3 Opaque grains

Scattered throughout the section are opaque grains, appearing black in both PPL and XPL at all rotations. These occur as elongate, skeletal, and blocky forms, in places wedged between the

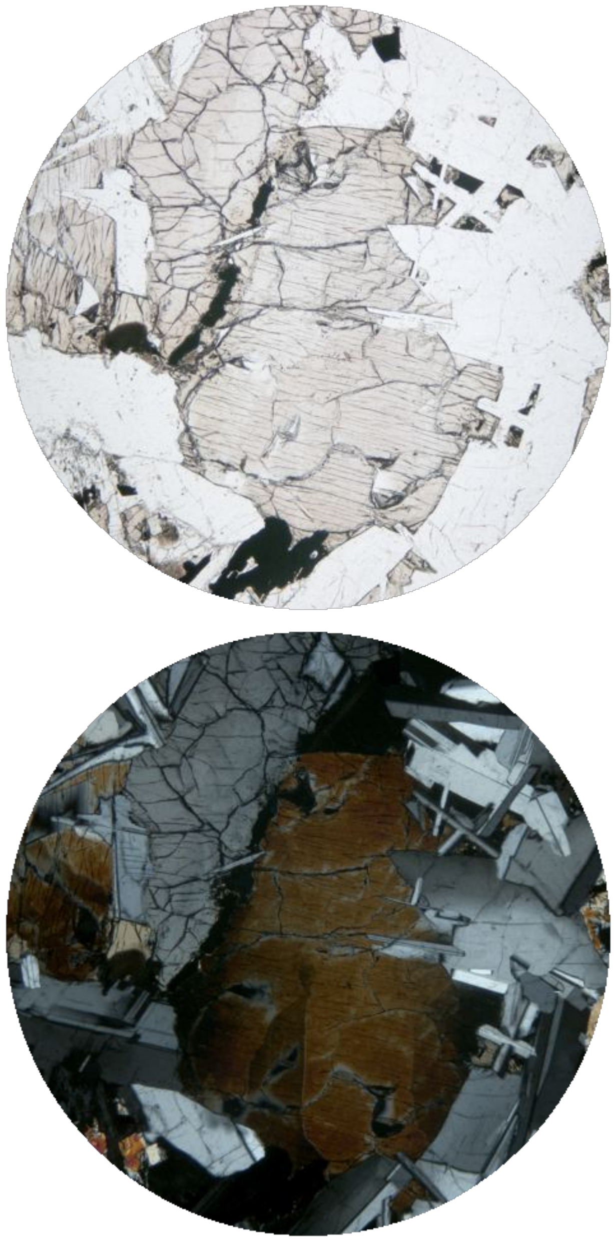

Figure 1: Sample 10047-r2 in PPL (top) and XPL (bottom). Field of view 2.0 mm

silicate phases. As seen in Figure 2, these grains appear bright in reflected light. They are likely ilmenite. The less reflective opaque grains may represent minor troilite and ulvöspinel.

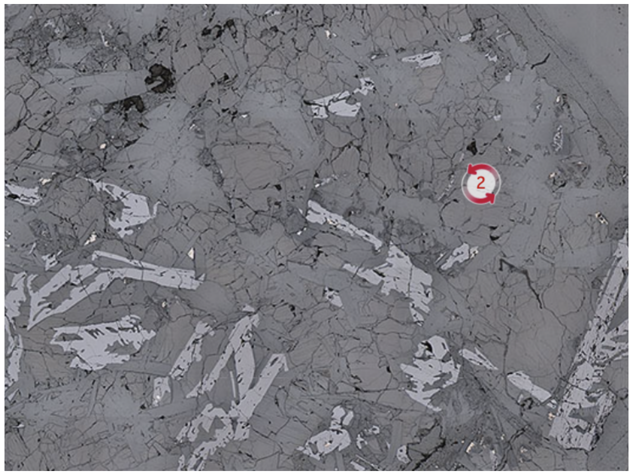

Figure 2: Sample 10047 under reflected light

2.4 Overall texture and Interpretation

The defining textural relationship in this section is ophitic to subophitic: euhedral to subhedral plagioclase laths are partially to completely enclosed within large, optically continuous pyroxene crystals. This geometry indicates the order of crystallization. Plagioclase and ilmenite nucleated early and grew as discrete laths and blades suspended in the melt, while pyroxene crystallized around them in large oikocrysts, engulfing the earlier-formed crystals as it grew.

Some pyroxene grains reaching several millimeters indicates relatively slow cooling [3]. The interstitial spaces between the major phases are where the residual melt solidified.

3 Segmentation of Opaques

You may want to give it a try to segment opaque phases in the PPL and XPL images, allowing the texture of the sample to be visualized more clearly: `https://huggingface.co/spaces/pingwang2026/segmentation-of-opaques-in-petrographic-images`

References

[1] "10047 Ilmenite basalt," *The Virtual Microscope*, The Open University. [Online]. Available: `https://www.virtualmicroscope.org/content/10047-ilmenite-basalt`

[2] C. Meyer, *Lunar Sample Compendium*. Houston, TX, USA: NASA Johnson Space Center, 2011. [Online]. Available: `https://curator.jsc.nasa.gov/lunar/lsc/`

[3] Ross, M., Bence, A. E., Dwornik, E. J., Clark, J. R., and Papike, J. J. (1970). Mineralogy of the lunar clinopyroxenes, augite and pigeonite. *Proceedings of the Apollo 11 Lunar Science Conference*